

Tawala Frontend

Theme Consistency & UX Engineering Audit

Canonical Modern Retail Operating System (DESIGN.md)

Branch: chore/theme-retail-os | Date: 2026-09-13

Method: Full frontend scan + UX Engineering AUDIT mode | Evidence from repository only

Scope: frontend/src routes, feature modules, hardcoded visual values, shared component adoption, information hierarchy, friction, recovery, and flow integrity against locked canonical tokens and `@/lib/components/ui`.

Out of scope: Backend correctness, production metrics, formal user interviews.

1. Executive summary

Product UI still largely uses the previous Silk & Slate language (Tailwind slate/emerald utilities, large radii, marketing-weight uppercase chrome). Canonical Modern Retail OS tokens and the shared UI kit exist and are proven on /themetest, but **no product feature module imports the shared kit**. That gap drives most theme inconsistency.

UX coverage is strong by domain (org, terminal, inventory, customers/credit, staff), yet frontline flows still expose jargon, dense hierarchy, and inconsistent empty/error treatment. Highest leverage: enforce AGENTS.md rules, migrate shells + POS to the kit, simplify checkout/credit hierarchy and stock-audit copy.

Severity model

Sev	Meaning	User consequence
P0	Blocks task or destroys trust	Cannot complete sale/credit/login; wrong money state
P1	Serious friction or hierarchy failure	Slow, error-prone, or confuses primary action
P2	Inconsistency / secondary confusion	Looks untrustworthy or trains bad habits
P3	Nice-to-have alignment	Minor cognitive load

Headline findings

- **P1 Theme dual system:** Tokens in globals.css; product UI hardcodes slate-*, emerald-*, bg-white, rounded-xl/2xl across org, staff, stock, sales, customers.
- **P1 Shared kit unused:** Only themetest imports @/lib/components/ui. Product controls are one-off class strings.
- **P1 Hierarchy drift:** Uppercase micro-labels and font-black marketing weight fight calm retail OS tone.
- **P1 POS/checkout load:** Terminal, cart, checkout, complete-sale chain is powerful but dense under queue pressure.
- **P2 Empty/error inconsistency:** Mixed tailored empties vs raw errors vs blanks.
- **P2 Nav IA:** Org vs business dual nav is correct; active states and labels are inconsistent.

2. Theme consistency audit

2.1 Locked canonical definition

Primary petrol #003F4E | Secondary terracotta #C1705B | Success mint #0F766E | Warm alabaster surface gradient | Radius 0.5rem structural / 1rem modals / 1.5rem pills | Border-first cards | Tabular money | Plus Jakarta + Inter | 48px primary targets.

2.2 Inconsistency evidence (scan)

Area	Legacy density	Examples
Legal / marketing	Very high	privacy, terms, solutions
Staff	High	TeamDirectory, StaffMemberWorkspace
Org shell / home / settings	High	OrgCommandCenter, OrgHome, Sidebar
Stock / inventory audit	High	AuditForm, StockTakingTableRow
Sales / cart / checkout	Medium	Mix of brand-* and ad-hoc radii
Customers / credit	Medium	CustomersList, CollectCreditForm
Shared kit in product	None	Only /themetest

2.3 Anti-patterns

- **Hardcoded palette:** bg-slate-*, text-slate-*, bg-emerald-600, bg-red-50, bg-white instead of bg-card / bg-register / text-muted / border-border / bg-brand-*.
- **Radius drift:** rounded-xl / rounded-2xl vs DESIGN.md 0.5rem structural default.
- **Type weight:** font-black + uppercase tracking-widest on operational controls.
- **One-off controls:** Raw button/input with long classes; focus rings and heights diverge.
- **Dark forks:** Many dark:slate-* while product default is light.
- **Success split:** emerald-600 vs token brand-accent / success for settlement.

2.4 Already aligned

- globals.css canonical tokens, @theme mapping, body surface gradient.
- Root layout fonts (Plus Jakarta Sans + Inter + mono).
- Some sales/terminal paths use brand-primary / card-layered / border-border.
- /themetest proves the full kit (forms, calendar, charts, table, modal, badges).

3. UX experience by domain

Users (from product context): Kenyan SME owners and cashiers. Jobs: sell fast, stop stock leakage, collect credit, know if the day is healthy, manage staff and branches.

3.1 Auth & onboarding

Flows: login, onboarding personal/org/plans/password, invite accept, forgot/reset. Friction: multi-step structure is sound; visual language still SaaS-marketing; invite vs self-serve can feel like two products. **Sev P2.**

3.2 Organization shell (HQ)

Flows: /org home, stores, staff, billing, settings, enter business. Friction: command-center density and slate styling hide the single next action (add branch vs invite staff vs billing). Paywall messages not always adjacent to the blocked action. **Sev P1.**

3.3 Terminal / POS

Flows: terminal, cart, checkout, complete-sale, staged recovery, out-of-stock filter, modifier-click remove. Strengths: power-user affordances, skeletons, cart sidebar on large screens. Friction: multi-route chain increases wayfinding under queue pressure; payment + customer + credit compete on checkout; money totals must always win the eye (not yet universal). **Sev P1 hierarchy; watch P0 if staged recovery fails silently.**

3.4 Inventory & stock

Friction: audit/restock jargon (e.g. Theoretical Expected System Stock) is system language, not shop-owner language. Dense uppercase labels. Strong slate-legacy visuals. **Sev P1 copy + P2 theme.**

3.5 Customers & credit

Strengths: list to detail IA; open-credit filter matches the job. Friction: collect-credit success should mirror mint settlement banners; mixed styling vs kit. **Sev P2.**

3.6 Staff

Friction: functional tables with pre-canonical visuals; emerald CTAs diverge from petrol primary; role assignment heavy for first-time owners. **Sev P2.**

3.7 Cross-cutting

- Primary action not always the strongest control.
- Destructive actions not consistently terracotta-coded.
- Loading: Loader2 vs pulse vs blank — no single Spinner/Skeleton language in product.
- Focus rings and 48px targets not guaranteed on ad-hoc buttons.
- Marketing pages may stay distinct but must not leak into app chrome.

4. Recommendations and fix plan

4.1 Non-negotiable rules (also in AGENTS.md)

- No hardcoded hex or Tailwind palette colors (slate/indigo/emerald/blue/gray/zinc) in product UI — semantic tokens only.
- No one-off buttons/inputs/selects/toggles/modals/badges/spinners in product features — use `@/lib/components/ui`.
- Money uses tabular/amount classes; settlement uses brand-accent / SuccessBanner.
- Destructive/void = secondary (terracotta); command = primary; pay = success.
- New tokens require DESIGN.md + globals.css + themetest updates together.

4.2 Phased remediation

Phase	Work	Outcome
A	AGENTS.md rules; stop new slate/emerald debt	Stop bleed
B	OrgShell, Sidebar, business layout chrome to tokens + kit	HQ canonical
C	Terminal, Cart, Checkout, CompleteSale to kit; money hierarchy	Frontline trust
D	Customers list/detail/collect + inventory/stock copy + kit	Credit and stock clarity
E	Staff directory/workspace + settings forms	Admin consistency
F	Public marketing isolate or restyle last	No false coupling

4.3 Copy and flow fixes

- Stock audit: System qty, Counted qty, Difference, Why? — not theoretical system discrepancy mapping.
- Checkout order: Total, Pay method, Customer if credit, Confirm; collapse secondary on mobile.
- Org home: one recommended next action when stores=0 or staff=0 or trial ending.
- Paywall blocks always paired with upgrade action in the same viewport.
- Empty states: one sentence + one primary Button from the kit.

5. User flow sketches (target)

Task language after remediation — not internal route maps.

5.1 Frontline sale

```
[Cashier on Terminal]
-> Search / tap products (stock > 0 by default)
-> Ticket updates (qty, line total, grand total always visible)
-> Charge
    -> Payment method (Cash | M-Pesa | Credit)
    -> If Credit: pick/create customer (required)
    -> Confirm
-> Settlement banner (matched / recorded)
-> New sale OR Print/SMS receipt
```

5.2 Collect credit

```
[Owner / cashier]
-> Customers -> filter Open credit
-> Open customer -> Collect
-> Amount (default balance, editable) + method
-> Confirm
-> Success banner + updated balance
-> Back to customer overview
```

5.3 Stock count

```
[Owner]
-> Inventory -> product -> Count
-> See system qty -> enter counted qty
-> See difference (plain language)
-> Optional reason if difference != 0
-> Save
-> History line written; list shows new qty
```

5.4 First-run org

```
[New owner after onboarding]
-> Org home shows ONE primary: Add your first branch
-> Create store -> Open terminal
-> Optional: Invite staff / Add products
-> Billing limits only when approaching or blocked
```

6. Wireframes — pages to amend first

Low-fidelity structure. Controls map to shared kit only.

6.1 Org home

```
+-----+
| Org nav: Home | Branches | Staff | Billing | Settings |
+-----+
| {Org name}                                         |
| [Primary: Add branch]      secondary: Invite staff |
+-----+
| Usage meters      | Branch list      | Alerts      |
| plan limits      | name / open      | trial / paypal|
+-----+
Kit: Button, Card, Badge, Table
```

6.2 Terminal

```
+-----+
| Search [      ] Filters      | Ticket      |
| Product grid (Cards)         | lines + qty |
| name price stock badge      | TOTAL amount-lg |
|                               | [Charge success] |
+-----+
Mobile: sticky bottom charge sheet
Kit: Input, Card, Badge, Button(success), Spinner
```

6.3 Checkout

```
+-----+
| Back      Checkout      |
| TOTAL      amount-lg    |
| Pay method Cash | M-Pesa | Credit      |
| Customer search (required if Credit)    |
| [Confirm payment – Button success]      |
+-----+
Kit: Button, Input, Select, Card
```

6.4 Customers list

```
+-----+
| Customers [Search] [Open credit Switch] [Add Button] |
| Table: Name | Phone | Open credit | Lifetime      |
| Empty: No customers yet + Button Add                |
+-----+
Kit: Input, Switch, Button, Table, Badge, Skeleton
```

6.5 Collect credit

```
+-----+
| Customer name      Open balance amount-lg    |
| Collect amount (currency Input)  Method Select |
| [Collect – Button success]      |
| SuccessBanner on complete      |
+-----+
```

6.6 Stock count

```
+-----+
| Product name      |
| System qty (tabular)      Counted qty Input |
| Difference (Badge success/warning)      |
| Reason Select if difference != 0      |
| [Save – Button primary]      |
+-----+
```

7. Hi-fi specification (canonical)

Hi-fi equals the token system plus kit. Amended pages must match /themetest canonical board. No parallel style sheets or hardcoded values.

7.1 Color application

Role	Token / component	Use
Page floor	body surface gradient	Global; never bg-slate
Card / panel	Card / bg-card + border-border	Content pods
Recessed well	bg-register	Table headers, wells
Primary action	Button primary	Command
Destructive	Button secondary	Void / delete
Pay / settle	Button success	Charge, collect
Money	amount-lg / amount-md / tabular	All KSh figures

7.2 Type, motion, responsive

Headlines via text-h1/h2/h3 (Plus Jakarta). Body via Inter. No third display font. Avoid font-black + uppercase tracking-widest on operational UI. Spinner/Skeleton for waits; toast or SuccessBanner for money completion; Modal for destructive confirm. Breakpoints and spacing from globals only; POS split $\geq 1024\text{px}$; sticky charge $< 768\text{px}$.

8. Priority matrix — pages to amend

#	Page / surface	Why
1	Terminal + Cart sidebar	Frontline revenue path
2	Checkout + Complete sale	Money confirmation
3	Org shell + Sidebar + Org home	Daily HQ entry
4	Customers list + Collect credit	Credit recovery job
5	Inventory + stock audit/restock	Leakage control + jargon
6	Staff directory + member workspace	Admin consistency
7	Business settings / store form	Secondary
8	Public marketing / legal	Isolate or restyle last

9. Acceptance criteria for any UI PR

- Only semantic tokens from globals.css (no slate-/emerald-/hex in product features).
- Interactive controls from @/lib/components/ui unless PR documents an exception.
- Primary task identifiable in 3 seconds; money figures tabular.
- Loading, empty, and error states for the happy path failure modes.
- Destructive actions use Modal confirm and secondary variant.
- npm run lint and npm run build pass; /themetest remains visual reference.

End of audit. Implementation of remediation phases requires explicit Engineer Mode approval. Recommended start: Phase A (rules — this PR) then Phase C or B by product priority.